

Instructions to the Students:

1. All the questions are compulsory
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question
3. Use of non-programmable scientific calculators is allowed
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

Q.1 Attempt Any Two

- A) Draw stress-strain curve for mild steel. Explain characteristic points of the curve. (CO1) 6
- B) A rod of steel is 10 meters long at a temperature of 20 degrees Celsius. Find the free expansion of the rod when the temperature is raised to 65 degrees Celsius. Find the temperature stress produced when the expansion of the rod is prevented. Take $E = 200 \text{ GPa}$ and $\alpha = 12 \times 10^{-6}$ per degree Celsius (CO2) 6
- C) A plane element in a boiler is subjected to tensile stresses of 400 MPa on one plane and 150 MPa on the other at right angle. Each of the above stresses is accomplished by a shear stress of 100 MPa such that when associated with the minor stress tends to rotate the element in anticlockwise direction. Find; (CO3) 6
- i. Principal stresses and their direction
 - ii. Maximum shear stress

Q.2 Attempt Any Two

- A) A wagon weighing 20 kN is attached to a wire rope and is moving at the speed of 5.4 kmph. The rope suddenly jams and wagon is brought to rest. If length of rope is 50 m and diameter is 36 mm, find maximum instantaneous stress and elongation of rope. Take $E = 200 \text{ GPa}$. (CO2) 6
- B) A rectangular pier is subjected to a compressive load of 450 kN as shown in figure 02. Find stress intensities on all the four corners of the pier.

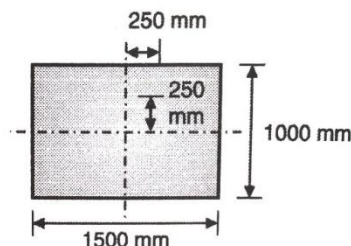


Figure 02

- C) Interpret a relationship between rate of loading, shear force and bending moment (CO4) 6

Q.3 Attempt Any Two

- A) An I-section beam 350 mm X 150 mm has a web thickness 10 mm & flange thickness 20 mm. If the shear force acting on the section is 40 kN, then find; (CO3) 6
- a) Maximum shear stress developed in the section
 - b) Sketch the shear stress distribution diagram
 - c) Total shear force carried by web

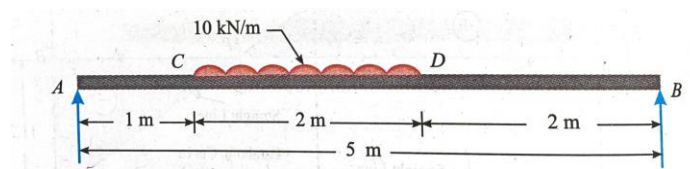
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|----|---|-------|---|
| B) | Derive an expression for Flexural equation along with assumptions | (CO2) | 6 |
| C) | Define the neutral axis in the context of pure bending. Discuss how it relates to the distribution of bending stress. | (CO3) | 6 |

Q.4 Attempt Any Two

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|----|--|--------------|---|
| A) | Derive an expression for Torsional formula along with assumptions | (Understand) | 6 |
| B) | A hollow shaft is to transmit 200 kW at 80 r.p.m. If the shear stress is not to exceed 60 MPa and internal diameter is 0.6 of the external diameters, find the diameters of the shaft. | (Apply) | 6 |
| C) | A hollow CI column of external diameter 200 mm, length 4 meter with both the ends fixed, supports an axial load of 800 kN. Determine the thickness of the column required by using Rankine's formula taking constant of $1/6400$ & working stress at 80 MN/m^2 | (Apply) | 6 |

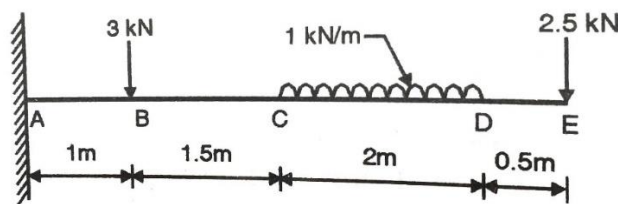
Q.5 Attempt Any Two

- A) Draw shear force and bending moment diagrams for the beam loaded as shown in figure below



(CO4) 6

- B) Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 04



(CO4) 6

Figure 04

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|----|--|-------|---|
| C) | Draw the shear force diagram and bending moment diagram for a cantilever beam carrying a point load at its free end. | (CO4) | 6 |
|----|--|-------|---|

***** Best of Luck *****